

Proof. **Degenerate case.** If $\sum_{i=1}^n b_i^2 = 0$, then each $b_i = 0$. Hence

$$\sum_{i=1}^n a_i b_i = 0, \quad \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) = 0,$$

so $0^2 = 0 \leq 0$. The equality condition is satisfied because the zero vector is proportional to any vector.

Non-degenerate case. Assume

$$A := \sum_{i=1}^n b_i^2 > 0$$

and define, for a real parameter t ,

$$Q(t) := \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is non-negative, therefore

$$Q(t) \geq 0 \quad \text{for all } t \in \mathbb{R}. \quad (1)$$

Expanding $Q(t)$ and using $(x - y)^2 = x^2 - 2xy + y^2$ gives

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= \underbrace{\sum_{i=1}^n a_i^2}_C - 2t \underbrace{\sum_{i=1}^n a_i b_i}_{S_{ab}} + t^2 \underbrace{\sum_{i=1}^n b_i^2}_A. \end{aligned}$$

Thus $Q(t) = At^2 + Bt + C$ with

$$A = \sum_{i=1}^n b_i^2 > 0, \quad B = -2S_{ab}, \quad C = \sum_{i=1}^n a_i^2.$$

A quadratic polynomial with leading coefficient $A > 0$ that satisfies $At^2 + Bt + C \geq 0$ for every real t must have a non-positive discriminant:

$$B^2 - 4AC \leq 0.$$

Applying this to (??) we obtain

$$(-2S_{ab})^2 - 4 \left(\sum_{i=1}^n b_i^2 \right) \left(\sum_{i=1}^n a_i^2 \right) \leq 0.$$

Dividing by $4 > 0$ yields

$$S_{ab}^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

i.e.

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

This is the Cauchy–Schwarz inequality.

Equality condition. Equality occurs precisely when the discriminant is zero, i.e. $B^2 - 4AC = 0$. Then the quadratic has the unique real root

$$t_0 = -\frac{B}{2A} = \frac{S_{ab}}{\sum_{i=1}^n b_i^2}.$$

Since $Q(t_0) = 0$ and each term $(a_i - t_0 b_i)^2$ is non-negative, we must have $a_i - t_0 b_i = 0$ for every i ; thus $a_i = \lambda b_i$ with $\lambda := t_0$. Conversely, if such a scalar λ exists, then every term $(a_i - \lambda b_i)^2$ vanishes, so $Q(\lambda) = 0$ and the discriminant is zero, giving equality in the Cauchy–Schwarz inequality. This also covers the degenerate case where all $b_i = 0$ (or all $a_i = 0$), because any λ satisfies $a_i = \lambda b_i$.

Hence, for all real families $(a_i)_{i=1}^n$ and $(b_i)_{i=1}^n$,

$$\boxed{\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)}$$

with equality iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (including the case where one of the vectors is the zero vector). $\square$